

LFEA Load and Equipment Interface Concept Note

Full 3D linear piping analysis for thermal, guide, line-stop, anchor, spring and nozzle loads

Repository baseline main @ 6ca233bf	Core basis LFEA B-2.x to B-4.0	Primary code ASME B31.3	Initial physics Linear static 3D
Mission: extend the existing LFEA chain into a production load-and-interface analysis workflow in which thermal forces, guide/line-stop reactions, anchor six-component loads and equipment nozzle loads are recovered from one governed 3D stiffness solution—never from tributary or empirical formulas.			

1. Purpose

This concept note defines the LFEA route for quantities that the first-cut screening method intentionally does not calculate. It covers full-model load cases, support and equipment-interface stiffness, solved displacements and reactions, nozzle-load assessment, B31.3 stress evaluation, traceable UI, reporting and qualification tests.

2. Existing Repository Foundation

Authority	Current role
B-2.1 to B-2.5	Mechanical-model contract, material state, pipe section, local axes and model compilation.
B-3.0	Immutable physical load cases: gravity, distributed weight, pressure, temperature, nodal force/moment, distributed load, equivalent static and prescribed movement.
B-3.1	3D frame element stiffness, equivalent loads, rigid offsets and transformations.
B-3.2	Piping component flexibility and component-owned code stations.
B-3.3	Global assembly, constraints, solve and reactions.
B-3.4	Element end actions, force fields, code-point resultants and envelopes.
B-4.0	B31.3 sustained, occasional and displacement-stress-range evaluation using caller-supplied datasets.
PR #14	Engineering SVG, Sequential Sketcher and split SVG/WebGL workbench surface.

3. Scope

3.1 Included in Initial Production Scope

- 3D small-displacement linear static piping analysis.
- PIPE, BEND, TEE/branch, REDUCER, finite rigid bodies and rigid links.
- Gravity, operating contents, insulation and component mass.
- Uniform element temperature change using the declared B-3.0 thermal-strain profile.
- Pressure state with separately authorized code stress, pressure stiffening, axial thrust and Bourdon flags.
- Nodal forces/moments, distributed loads, wind/seismic equivalent-static loads and prescribed movements.
- Anchors, guides, line stops, bilateral rests, linear springs and six-DOF equipment/nozzle stiffness.
- Signed support and nozzle actions in global and declared local coordinate systems.
- B31.3 sustained, occasional and displacement-stress-range assessment.

3.2 Explicitly Deferred

- Automatic gaps, lift-off, one-way contact and friction.
- Nonlinear springs, constant-effort hanger mechanics and load-travel design selection.
- Dynamic response spectrum, time history, water hammer and acoustic vibration.
- Pipe-soil contact and buried-pipe nonlinear behavior.
- Plasticity, large displacement and local shell/nozzle flexibility FEA.
- Temperature-gradient bowing across a pipe section.

4. Governing Architecture

Engineering source / imported piping JSON

- > conditioned 3D topology
- > material, section and local-axis resolution
- > support and nozzle interface compilation
- > fea-linear-model/v1
- > B-3.0 physical load cases
- > B-3.1 / B-3.2 element and component contributions
- > B-3.3 sparse assembly and linear solve
- > B-3.4 displacement, reaction and component recovery
- > support-load and nozzle-load envelopes
- > B-4.0 B31.3 assessment

-> immutable evidence, SVG/WebGL review and reports

4.1 Authority Separation

- B-3.0 declares physical loads and authorizations; it does not create element load vectors.
- B-3.1/B-3.2 create stiffness and equivalent nodal actions.
- B-3.3 solves and owns nodal reactions.
- B-3.4 recovers local/global element and interface actions.
- B-4.0 evaluates code stress from recovered actions; it does not solve.
- Workspace renderers consume sealed results and perform no mechanics.

5. Engineering Model Extensions

5.1 Support Interface Contract

fea-linear-support-interface/v1

supportId
attachedNodeId / attachedComponentStation
supportFrame
restraintComponents:
 UX UY UZ RX RY RZ
mode per component:
 FREE | FIXED | LINEAR_SPRING | PRESCRIBED
stiffness values and units
rigid offset / support elevation
physicalDirectionLabels:
 VERTICAL | LATERAL | LONGITUDINAL | ROTATIONAL
sourceEvidence
semanticHash

- Guide and line-stop definitions require an exact support local frame.
- Pipe tangent supplies longitudinal direction; vertical comes from the project coordinate profile; lateral is the orthogonal completion.
- Support offsets must transfer forces and moments to the physical attachment/reference point.
- Bilateral linear restraints may recover tension; when the real support is unilateral, the result is CONDITIONAL and must be checked by a future contact model.

5.2 Equipment Nozzle Interface Contract

fea-linear-equipment-nozzle/v1

equipmentId
nozzleId
interfaceNodeId
nozzleFrame
connectionPoint
rigidOffset
stiffnessMode:
 RIGID
 DIAGONAL_6DOF
 FULL_6X6
 PRESCRIBED_MOVEMENT
stiffnessMatrix / prescribedValues
allowableProfileRef
sourceEvidence
semanticHash

5.3 Nozzle Stiffness Modes

Mode	Use
RIGID	Nozzle node fixed to equipment reference. Appropriate only when equipment flexibility is intentionally neglected.
DIAGONAL_6DOF	Independent translational and rotational stiffness values. No coupling.
FULL_6X6	Symmetric coupled stiffness matrix. Requires unit, symmetry and positive-semidefinite qualification.
PRESCRIBED_MOVEMENT	Equipment thermal growth, settlement or vendor movement imposed at the nozzle interface.

6. Physical Load Cases

6.1 Gravity and Mass

- Pipe wall, contents and insulation produce distributed weight.
- Valves, flanges, instruments and other components use explicit point mass/COG or finite-rigid-body mass.
- EMPTY, OPE and HYD remain independent physical cases.
- Gravity vector and acceleration are declared in the load-case profile.

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- No mass defaults or double counting are permitted.

6.2 Temperature and Thermal Strain

Initial supported rule:
 $\epsilon_{thermal} = \alpha_{evaluation} * (T_{case} - T_{reference})$

Element initial-strain load:
 $f_{thermal_local} = \int (B^T D \epsilon_{thermal} dV)$
represented by the qualified frame/component formulation

Required evidence:
reference temperature
element evaluation temperature
alpha material state
temperature source and revision
thermal approximation profile

- Uniform temperature may vary by element or component zone.
- Temperature-dependent alpha integration is a later compiler, already identified as deferred by B-3.0.
- Thermal force is never computed as $E A \alpha \Delta T$ unless that closed-form benchmark exactly matches the structural restraint condition.
- Guide, line-stop, anchor and nozzle thermal loads are recovered from the solved system.

6.3 Pressure

Authorization flag	Meaning
codeStress	Pressure contribution used by B-4.0 code evaluation.
pressureStiffening	Optional formulation effect; only when implemented and qualified.
axialThrust	Explicit closed-end/reducer/valve pressure thrust; topology and boundary evidence required.
bourdon	Optional pressure-deformation effect; separately qualified.

A pressure value alone authorizes none of these effects. Each flag is declared and proven downstream.

6.4 Applied and Occasional Loads

- Nodal forces and moments.
- Uniform or linear distributed loads in global or element-local basis.
- Equivalent-static wind and seismic loads with explicit direction, factor and source.
- Prescribed anchor/nozzle movements and support settlement.
- Static relief, slug or other event loads only as explicit equivalent-static input; dynamic qualification remains outside scope.

7. Support Load Evaluation

7.1 Guide Loads

A guide load is the solved reaction component in the support local lateral direction. It is not a percentage of weight. The result retains sign, case, support frame, physical attachment point, offset moment and recovery hash.

7.2 Line-Stop Loads

A line-stop load is the solved reaction in the local longitudinal pipe direction. It typically arises from thermal expansion, prescribed movements, occasional loads and pressure effects where explicitly authorized. No empirical section-modulus formula is used.

7.3 Anchor Loads

Anchor result per physical load case:
Fx, Fy, Fz
Mx, My, Mz

Retain:
global basis
anchor local basis
action/reaction sign convention
attachment and reference point
rigid-offset transformation
governing case / envelope
equilibrium evidence

7.4 Rest and Spring Loads

- A bilateral linear rest produces a signed linear reaction. If tension is physically impossible, flag the case as CONTACT_REVIEW_REQUIRED.
- A linear spring reaction derives from solved displacement, declared stiffness and any governed preload representation.
- Cold load, hot load, travel and load variation are reported only when the spring model carries sufficient evidence.
- Snubbers are not automatically active in sustained/thermal cases; static-equivalent activation requires a declared occasional profile.

8. Equipment Nozzle Loads

8.1 Recovery

- Recover the six-component action at the nozzle interface from B-3.4 element/interface equilibrium.
- Transform from global to the declared equipment/nozzle coordinate frame.
- Retain both piping-on-equipment and equipment-on-piping sign senses with an explicit convention.
- Include rigid-offset moment transfer to the physical nozzle face or vendor reference plane.
- Fold min/max/absolute-max envelopes while retaining the governing physical load case.

8.2 Nozzle Allowable Assessment

No nozzle allowable values are embedded. Every allowable envelope is caller-supplied from vendor data, an authorized equipment standard/profile, or a project criterion.

Profile	Assessment authority
VENDOR_COMPONENTWISE	Independent allowable Fx/Fy/Fz/Mx/My/Mz values and case applicability.
VENDOR_INTERACTION	Caller-supplied linear, quadratic or custom interaction rule.
API_610_PROFILE	Pump nozzle-load profile supplied by the project/vendor dataset.
API_617_PROFILE	Compressor nozzle-load profile supplied by the project/vendor dataset.
NEMA_SM23_PROFILE	Steam-turbine load profile supplied by the project/vendor dataset.
PROJECT_PROFILE	Owner/project-specific allowable envelope.

8.3 Nozzle Assessment Record

lfea-nozzle-load-assessment/v1

equipmentId / nozzleId
loadCaseId / envelopeId
nozzleFrame
forces and moments
allowableProfileId and dataset hash
term-by-term ratios
interaction value and rule ID
status:
 QUALIFIED_UNDER_CONFIGURED_PROFILE
 CONDITIONAL
 BLOCKED
limitations
semanticHash / evidenceHash

9. B31.3 Stress Evaluation

- SUSTAINED uses qualified pressure contribution plus recovered sustained actions.
- OCCASIONAL uses recovered occasional actions and declared duration/category factors.
- DISPLACEMENT_STRESS_RANGE uses a case-pair/range action, not the sum of absolute operating results.
- OPERATING remains a result view, not a B31.3 compliance category.
- B31J SIFs and indices are caller-supplied and direction-specific.
- Code result hashes bind profile, edition dataset, factors, material, section, recovery and combination identity.

10. UI and Workflow

Model / Import
-> Supports & Nozzles
-> Load Cases
-> Preflight
-> Compile
-> Solve
-> Reactions
-> Nozzle Loads
-> B31.3
-> Export

Surface	Required behavior
Support table	Type, attachment, frame, restraint components, stiffness, offsets, source and readiness.
Nozzle table	Equipment/nozzle ID, frame, stiffness mode, prescribed movement, allowable profile and readiness.
Load-case manager	Gravity, temperature, pressure flags, applied loads, wind/seismic, prescribed movements.
Result viewport	Deformation, support vectors, anchor six-component glyphs, nozzle vectors, force/moment diagrams and code utilization.

Surface	Required behavior
Result inspector	Exact numeric values, basis, sign convention, parent hashes, formula/load evidence and governing case.
State-state control	Any engineering edit invalidates compilation, execution, recovery, nozzle assessment and code results.

11. Outputs and Exports

- Support-load CSV/JSON for civil and support design, with global/local actions and governing cases.
- Anchor-load summary with six components per case and envelope.
- Nozzle-load datasheet per equipment/nozzle and allowable assessment.
- Nodal displacement, element end actions, force/moment diagrams and code-point results.
- B31.3 code result tables with dataset and factor identities.
- Engineering report and immutable audit package.

12. Qualification Tests and Benchmarks

12.1 Thermal Benchmarks

Benchmark	Acceptance
TH-01	Free uniform-temperature bar: free expansion and zero reaction.
TH-02	Fully restrained bar: $E\alpha\Delta T$ reaction with independent closed form.
TH-03	Two-anchor straight pipe: thermal force and equal/opposite reactions.
TH-04	Planar U-loop: displacement and anchor loads against independent matrix/commercial comparison.
TH-05	Multiple temperature zones and material alpha states.
TH-06	Prescribed nozzle/anchor movement superposed with temperature.

12.2 Guide, Line-Stop, Anchor and Spring Benchmarks

Benchmark	Acceptance
SUP-01	Guide local-frame reaction and zero force in free components.
SUP-02	Line-stop axial reaction from thermal expansion.
SUP-03	Anchor cantilever six-component reaction against closed form.
SUP-04	Rigid support offset transfers force to correct moment.
SUP-05	Linear spring reaction, displacement and energy.
SUP-06	Bilateral rest tension flagged for contact review.
SUP-07	Global force and moment equilibrium for every solved case.

12.3 Nozzle Benchmarks

Benchmark	Acceptance
NOZ-01	Cantilever pipe connected to rigid nozzle; interface action equals fixed-end action.
NOZ-02	Nozzle local/global transformation and sign convention.
NOZ-03	Prescribed nozzle movement produces independent stiffness result.
NOZ-04	Diagonal 6DOF stiffness against closed-form spring response.
NOZ-05	Full 6x6 coupled stiffness; symmetry and energy checks.
NOZ-06	Rigid-offset moment transfer to nozzle face.
NOZ-07	Component-wise allowable assessment.
NOZ-08	Custom interaction rule and dataset-hash invalidation.
NOZ-09	Envelope retains governing case for each component.

12.4 Pressure and Code Benchmarks

- Pressure code-stress flag without thrust/stiffening/Bourdon.
- Axial-thrust authorized and blocked cases.
- Pressure-stiffening and Bourdon remain blocked until formulations are qualified.
- B31.3 sustained, occasional and displacement-range hand calculations.
- Operating category refusal and factor cross-application regressions.

12.5 Commercial Corroboration

Selected system benchmarks should be compared with one recognized commercial pipe-stress program such as CAESAR II, AutoPIPE or CAEPipe. The external program is corroborating evidence; closed-form and independently controlled benchmark values remain

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primary. Differences must be classified by formulation, component flexibility, support model, load definition, pressure effect, code profile or defect.

13. Implementation Work Packages

Package	Scope	Exit
LFEA-L0	Exact-main audit and interface authority	Confirm current B-3.0/B-3.3/B-3.4 contracts and reaction availability.
LFEA-L1	Support and project-frame contracts	Support local frames, offsets, six components and linear springs.
LFEA-L2	Equipment/nozzle contracts	Nozzle frames, stiffness modes, movements and allowable references.
LFEA-L3	Load compiler completion	Thermal zones, pressure flags, equivalent static and prescribed movement.
LFEA-L4	Assembly and reaction recovery	Support/nozzle contributions, six-component recovery and equilibrium.
LFEA-L5	Nozzle assessment	Vendor/API/NEMA/project profiles and interaction records.
LFEA-L6	Range and code completion	Case-pair recovery and B31.3 displacement-range integration.
LFEA-L7	Workbench and reports	Load-case manager, support/nozzle panels, viewport, exports and reports.
LFEA-L8	Qualification	Benchmarks, commercial corroboration, anti-drift and exact-head release.
LFEA-NL1	Future nonlinear contact	Gaps, lift-off, friction and nonlinear springs; separate authority.

14. Anti-Drift Controls

- Block workspace calculation of guide, line-stop, anchor or nozzle loads.
- Block percentage-of-weight and empirical section-modulus reaction formulas.
- Block pressure effects unless the corresponding B-3.0 authorization flag is explicit.
- Block nozzle allowable values embedded in source code.
- Block support/nozzle results without basis, sign convention, case and recovery hash.
- Block operating stress being labelled B31.3 compliance.
- Block automatic lift-off/friction in the linear package.
- Block stale reactions or nozzle assessments after source, stiffness or load-case changes.
- Block local-frame inference where tangent/vertical evidence is unresolved.
- Block renderer recomputation of loads, stress or utilization.

15. Completion Criteria

1. Every support and nozzle has a qualified attachment and coordinate frame.
2. Thermal, gravity, pressure, mechanical and prescribed-movement cases are immutable and traceable.
3. Guide, line-stop, anchor, spring and nozzle loads are recovered from B-3.3/B-3.4.
4. Global and local actions satisfy force/moment equilibrium and sign conventions.
5. Nozzle assessments use caller-supplied allowable profiles and retain governing cases.
6. B31.3 sustained, occasional and displacement-range results are recovery-bound.
7. UI and exports cannot present stale or screening values as solved reactions.
8. All analytical, system and commercial corroboration benchmarks pass at the exact PR head.

LFEA LOAD & EQUIPMENT INTERFACE PROGRAM: QUALIFIED only after LFEA-L0 through LFEA-L8 pass. Nonlinear contact remains NOT IMPLEMENTED until LFEA-NL1 is separately qualified.

References

- Advanced_Analysis main at merge commit 6ca233bfe8e6fd3ee120f1fe4a274e847538291d.
- LFEA B-3.0 physical load-case contract and pressure-effect authorization flags.
- LFEA B-3.4 result recovery and B-4.0 B31.3 code engine.
- ASME B31.3-2024 Process Piping and ASME B31J-2023.
- API equipment standards and vendor allowable datasets, used only through caller-supplied profiles.